ep4nn: Evidence-Gated Structural Growth in a Connectome-Compiled Equilibrium Network

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Abstract

A measured connectome supplies structure and no dynamics. We study what should make a network built on such a structure grow. The system, `ep4nn`, starts as the 302-neuron *C. elegans* wiring imported into a rate-based equilibrium model and appends neurons and synapses when it witnesses two situations that the world declares different but that produce nearly parallel answers at a declared readout. Growth passes three gates: a prerequisite gate that asks whether the answer moves at all, an evidence gate that requires more than one collision, and an outcome gate that restores the previous graph unless the confusion falls. We report measurements rather than claims. On one laptop CPU the library settles a graph of 10,000 neurons and 103,943 synapse classes in 6.48 ms and a graph of 100,000 neurons and 1,239,921 synapse classes in 77.4 ms, at about 1.76 ns per synapse. A state-dependent contraction certificate exists on the worm graph where the global worst-case bound fails: the local spectral radius is 0.529 with a positive weighted witness at 0.696, against a worst-case row bound of 88.0 that certifies no budget. The growth machinery behaves as specified: over 1200 ticks it refused growth 1200 times because the prerequisite failed, and an earlier variant without the prerequisite gate produced thirty growths that the verdict layer rolled back. The main negative finding is that prerequisite. Across sixty operating points and nine situations the declared readout responds along a single direction: the best spread is 0.0777 against a gate of 0.05, and the largest absolute cosine between two context answers that must differ is 1.000 at every point. Capacity growth is therefore not the binding constraint, and a growth study on this pair would measure an uninformative object. We conclude that structural growth studies must publish the prerequisite measurement alongside the growth result, and we give a receipt format that makes such claims comparable across connectome projects. All numbers in this paper are produced by the accompanying probes.

Keywords: structural plasticity; equilibrium networks; connectome; worm wiring; contraction certificate; negative results; reproducibility receipts.

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1 Introduction

A measured connectome is a graph of who contacts whom. It is not an operating point, not a dynamical model, and not a behaviour. Three years of open fly-connectome work demonstrate the gap concretely: a complete male central nervous system of about 166,000 neurons is public under a permissive license, yet the projects that run it either supply the controller, supply the semantic mapping from cell groups to behaviour, or supply the reward. The most disciplined attempt to remove the controller reached a negative result: driving the ventral nerve cord through two descending neurons produced a 45.5 Hz population oscillation that a scrambled wiring reproduced as well.

Our question is narrower than behaviour. Suppose a network starts small, on a measured wiring, and must *add structure* as the situations it faces accumulate. What should trigger growth, what should growth wire, and how do we know the growth was worth it?

We answer with a system and a set of measurements. The system, `ep4nn`, treats growth as a repair of one witnessable failure: two situations that the world declares different share the present input and produce answers that are nearly parallel at the declared readout. The repair is a low-dimensional generator that appends neurons wired from the cells that carry the difference, and a verdict that measures the same collisions again and rolls the growth back when they do not improve.

The main result of this paper is that the repair is blocked by a prerequisite we did not expect to be the binding constraint: on the worm graph, at sixty operating points, the declared readout responds along a single direction. Answers differ in magnitude, not in pattern, so every stage of growth measures a code that carries nothing a lesson can attach to. This is the same failure the library reports for the fly antennal lobe under one global gain, and it is detectable before any growth is attempted. We therefore state the prerequisite as a rule and report its measurement as the headline, together with the growth machinery that enforces it.

1.1 Contributions

1. **A growth contract with a prerequisite.** Three gates, in order: the prerequisite gate (does the declared readout move), the evidence gate (more than one collision), and the outcome gate (rollback unless the confusion falls). A refusal to grow is an explicit recorded outcome, and in the measured configuration it is the only outcome that occurs (1200 of 1200 ticks).

2. **A low-dimensional structural generator.** Thirteen genes decide how many neurons an event appends, where their edges come from, and whether they reach the readout. Partner choice is driven by the difference between the colliding answers and its one-hop neighbourhood, and the reverse edge is added with a probability so that the equilibrium contrast contract survives. The parameter count sits four orders below the graph the generator can touch, which is what separates structural plasticity from implicit weight learning.

3. **A state-dependent contraction certificate.** On a measured connectome the global worst-case bound is vacuous ($q = 88.0$, with 292 of 302 rows at or above one). Evaluating the same quantity with the activation slopes of the current state gives spectral radius 0.529 and a positive weighted witness $\alpha = 0.696$, hence a settling budget.

4. **Measurements of the substrate.** Settlement cost and transport behaviour on the worm graph and on sparse graphs to 100,000 neurons, a certificate comparison, and a negative control result for continuous tracking in a closed loop.

5. **A prerequisite measurement, negative, with a receipt format.** Sixty operating points, nine situations, one direction; and a six-field receipt that separates nominal from executed scale, so that the refusal is comparable with the runs of other projects.

1.2 What is not claimed

No growth was kept, so this paper makes no claim about the benefit of structural growth, only about the contract that would measure it and the prerequisite that currently blocks it. No claim is made that the worm graph is an unsuitable substrate in general: the negative result is specific to the declared readout, the rate model, and the scanned operating points. No gradient contract is claimed for directed wiring, and no behaviour, embodiment or sensory front end is claimed: the senses and the outcome are declared by the world, which the receipt records as supplied.

2 Related Work

Equilibrium learning. The network is settled by the contrast of a free and a nudged fixed point, the two-phase form of equilibrium propagation [14] and of contrastive Hebbian learning [12], with the learned quantity carried by local efficacies on edges. The library's own documentation states the boundary we rely on: reciprocal learning ties paired weights, and arbitrary directed imports need not satisfy that gradient contract. Equilibrium propagation scales to deep convolutional networks by reducing the bias of its gradient estimator [8], sits beside implicit differentiation and bilevel formulations [19], and shares its object with deep equilibrium models [1]. Predictive-coding rules approximate backpropagation along arbitrary computation graphs [11, 16]. None of these rules grew a graph.

Complementary memories and replay. Fast episodic storage with slow consolidation is the complementary learning systems proposal [10]. The library implements one version: records written once by day, fixed dreams at night taught to slow weights, and a residual written back at dawn. Its own measurements report held-out sign accuracy rising from 0.55 to about 0.76–0.80 on a Connect Four slice, at a night cost of roughly 140–210 s per 1000 rows for a 4096-cell store, with the limit that dreams reproduce the store's errors as faithfully as its regularities. We adopt the two-scale picture as the time-scale structure of the system but do not evaluate it here.

Fly connectomes and their controllers. The adult female brain reconstruction contains 139,255 neurons [5], and the male central nervous system, which includes the ventral nerve cord and therefore the leg and wing motor neurons, about 166,000 [9]. The proofread release used by the library's fly lane carries 150,802 neurons and 1,877,099 connection classes [2]. Open projects in this space differ mainly in what they supply. A browser-facing demonstration with fly cell-type names runs a hand-wired network of 850 neurons [7]; a learned consequence model is paired with an application-level controller [4]; a directly read motor output was studied offline with the reported outcome that real-time neural control was not demonstrated, because the driven cord oscillated at 45.5 Hz and a scrambled wiring reproduced the oscillation [17]; and an on-device implementation reports 10 Hz on a handset using sparse storage with per-group gating [6]. The library's own fly guide reports that at one global gain the antennal lobe ignites, that Kenyon cell codes for two odours reach a cosine of 0.98, and that a gain per cell class brings the code to 4 and 11 percent sparsity with a cosine of 0.32 [3]. We reuse two practices from these projects: protocols written before the run, and receipts that separate nominal from executed scale. The spiking model parameters used in the fly lane, and the observation that an explicit synaptic delay moves a runaway oscillation upward in drive frequency, are documented in [15, 18]; our rate model carries neither, which the receipt records.

Conditional computation. Mixture-of-experts routing, early exit and pruning vary the work done per input at the weight level. Dynamic structure methods vary the graph. What distinguishes the trigger here is that it is a witnessable indistinguishability at a declared readout, not a loss, a router score, or an offline search objective.

3 System

3.1 The four rules and the gate before them

Prerequisite. The declared readout is a named set of neurons, not the whole state. The whole state nearly always separates, because it contains the driven cells themselves. The system measures

$$\texttt{readout_spread} \;=\; \frac{1}{|R|}\sum_{r\in R}\Big(\max_{s} a_r(s) - \min_{s} a_r(s)\Big),$$

the mean over readout cells R of the range of activation across the declared situations s. Below `min_response` the system records `no_response` and does not grow: added capacity cannot separate answers that do not move, so growth would be the wrong repair and its recorded outcome would carry no information.

Rule 1, demand. Two situations that the world declares different, share the present input, and produce answers whose absolute cosine exceeds `code_cap` constitute a capacity signal. The

signal is a witness rather than a loss: it names the pair that is being confused, and that pair is what the generator wires against.

Rule 2, evidence. A single collision can come from a settling failure or a transient. Growth is attempted only after `min_evidence` collisions have been observed and `min_ticks_between` ticks have passed.

Rule 3, growth. Each event appends k neurons. Each new neuron draws in-edges and out-edges from a degree prior, prefers partners among the cells that carry the difference between the colliding answers and their one-hop neighbourhood, keeps the graph's excitatory fraction, adds the reverse edge with probability `reciprocal` so the equilibrium contrast contract is preserved, and projects into the readout with probability `to_readout`. The generator is low-dimensional by construction.

Rule 4, verdict. After growth the same collisions are measured again. Growth is kept only if the collision cosine falls by `improve_margin`. Otherwise the connectome and the efficacy array are restored from the snapshot taken before the attempt, and the attempt is counted as a failure.

3.2 The operating point

A connectome carries contact counts and transmitter signs. It does not carry excitability, and the raw counts of the worm graph give row sums in the hundreds at which the rate model does not settle. The system declares an operating point by row-normalising the signs to `row_sum`. The library offers two further instruments for the same purpose, a bias calibration and a selected gain; Section 5.4 measures all three.

3.3 A settling budget from a state-dependent certificate

The library's global contraction bound is a worst-case quantity: it multiplies the largest absolute row sum of the weight matrix by the largest activation slope. On a measured connectome most neurons are inactive at the settled state, so the bound is far larger than the actual contraction. We evaluate the same object with the slopes of the current state,

$$K \;=\; (1 - \mathrm{d}t)\, I \;+\; \mathrm{d}t\, W \,\mathrm{diag}\big(s'(v)\big), \tag{1}$$

and report its spectral radius together with a positive vector h satisfying $Kh \le \alpha h$. The settling budget follows as $k \approx \log(\epsilon/\delta)/\log(1/\alpha)$. The same quantity is the precondition for reading the free and nudged contrast as a gradient, which is why it is measured at run time rather than assumed.

Remark 1. The certificate is state-dependent and therefore not transferable: a network that moves to a different activity pattern must re-certify. We treat that as the honest cost of not carrying a global bound that does not apply.

4 Implementation

Data. The wiring is the openworm c302 `A_Full` release of the hermaphrodite [13]: 302 neurons, 2279 chemical synapse classes and 1084 gap junctions, the latter imported in both directions. After dropping autapses and merging parallel edges the graph has 3137 directed synapse classes, 10,133 contacts, a density of 3.44 percent, and 61 sensory, 91 interneuron, 118 motor, 10 command and 22 pharyngeal cells. The file's soma positions are used for nothing here. The committed copy carries SHA-256 `2a7b117f`.

Library reuse. The network is a connectome object with a brain on it, using the library's rate neuron model, its free and nudged phases, its residual qualification, and its sparse transport.

Table 1: Settlement cost. Row-normalised operating point, tolerance 10^{-3}, chunk 32. The worm graph needs five times as many sweeps as a random graph of the same size and mean degree. Source: `probes/probe_scaling.py`.

graph	neurons	synapses	in-deg	transport	μs/sweep	sweeps
worm, measured	302	3137	10.4	blocked	27.2	160
sparse random	1000	10,337	10.3	blocked	161.4	32
sparse random	10,000	103,943	10.4	sparse	193.0	32
sparse random	100,000	1,239,921	12.4	sparse	2185.8	32

No new learning rule is introduced. The growth layer sits outside the library because the connectome is immutable and because the library forbids growth by design; a growth event rebuilds the connectome, which costs one pass over the edge arrays, and carries the learned efficacy across by regenerating the key order the library sorts by.

Cell classes. The connectome file carries a role for every neuron, which is the only cell class the wiring provides; the operating-point scan uses those roles to define the classes it gains. This mirrors the library's fly practice of a gain per cell type, with the difference that the worm's classes are coarse.

Receipt. Every run writes a JSON receipt with six fields: dataset with a hash; nominal against executed counts with their ratio; the dynamical model, its parameters, and what is absent (delay, adaptation, heterogeneous cell types); whether the loop is closed and which parts are supplied; the premise measurements; and the cost. The receipt is written by the run that produced it, so two results can be compared without reading anyone's prose.

5 Measurements

All measurements are from one Apple Silicon laptop, CPU only, single-threaded, Python 3.13 with NumPy 2.5 and SciPy 1.18, against the released library at version 0.80.0 plus local commits. Each table names the probe that produced it.

5.1 Settlement cost and scaling

Cost is about 1.76 ns per synapse, so the control frequency of a settling brain is bounded by

$$f \approx \frac{1}{n_{\text{syn}} \times 1.8\,\text{ns} \times \text{sweeps}}.$$

A 10,000-neuron sparse graph settles in 6.48 ms and a 100,000-neuron graph in 77.4 ms, which makes the first a real-time-scale object and the second an offline one. Two library defaults matter at this scale. The default constructors wire at about 30 percent density independent of size, measured at 29.1, 30.8, 31.7 and 32.3 percent at 152, 280, 536 and 1048 neurons, so 10,000 neurons would carry about 30 million synapses and settle in about 1.7 s. The transport path is chosen by a size test rather than by density, so a 3.4 percent dense graph pays a block array 29 times larger than its edge count. On the worm graph the sparse path is 1.78 times faster end to end (554 μs against 987 μs per settle), while on a 32 percent dense graph the blocked path is faster; the choice must therefore follow density. A batch size above one builds a temporary array of the full edge count, so a closed control loop should keep batch one.

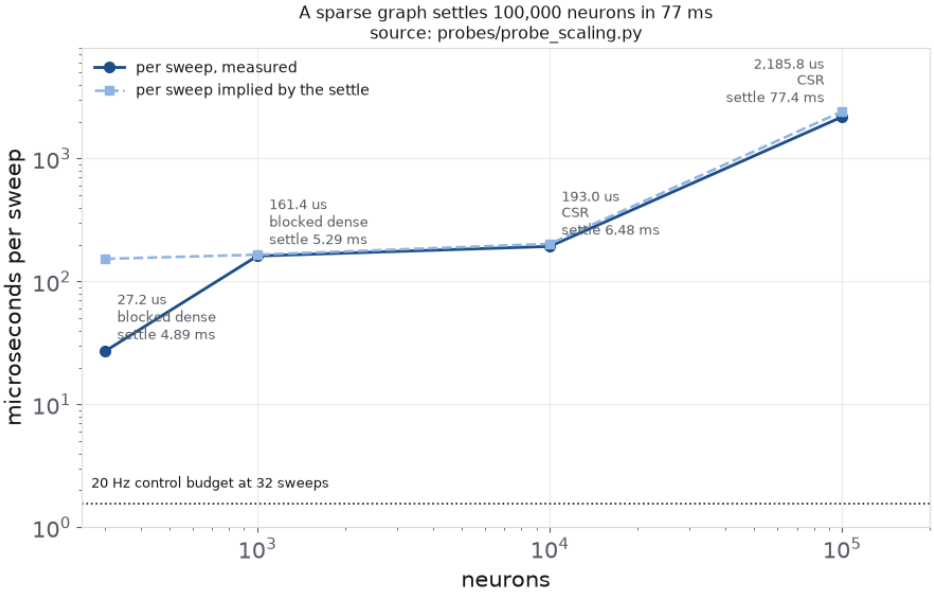


Figure 1: Cost per sweep against size on a log-log axis, and the budget for 20 Hz control at 32 sweeps. The worm graph is five times more expensive per sweep than a random graph of the same size and degree, which locates the difficulty in the spectrum rather than in the size.

Table 2: Contraction on the worm graph at the declared operating point. The global bound is vacuous; the state-dependent certificate is not. Source: `probes/probe_certificate.py`.

quantity	value
global worst-case bound q	88.0 (292 of 302 rows at or above 1)
local bound, current state	1.186 (1 of 302 rows)
spectral radius of K in (1)	0.529
weighted witness α, $Kh \leq \alpha h$	0.696 ($h \in [2.89, 11.08]$)
active neurons at the settled state	294 of 302

5.2 A certificate where the global bound fails

The global bound gives no certificate at this gain; the state-dependent one does, and it converts a refusal into a bounded settling budget. Two warnings follow from the same measurement. First, a certificate is a property of a state, not of a graph. Second, the bound is a sufficient condition; its failure at a particular gain is not evidence that settling fails, only that the corresponding guarantee is unavailable.

5.3 Behaviour of the growth machinery

In both variants the brain never shrank, every synapse kept an efficacy across every rebuild, the readout stayed inside the current graph, and the self-check passed. In the first variant the collision cosine read 1.000 before and after every attempt, so the verdict layer rolled all thirty attempts back at about one millisecond of rebuild each. In the second the prerequisite measured 0.019 and the system declined to act, recording why. Two failures of the machinery were found and fixed during construction, and both are worth stating because they are cheap to reproduce: a calibration that

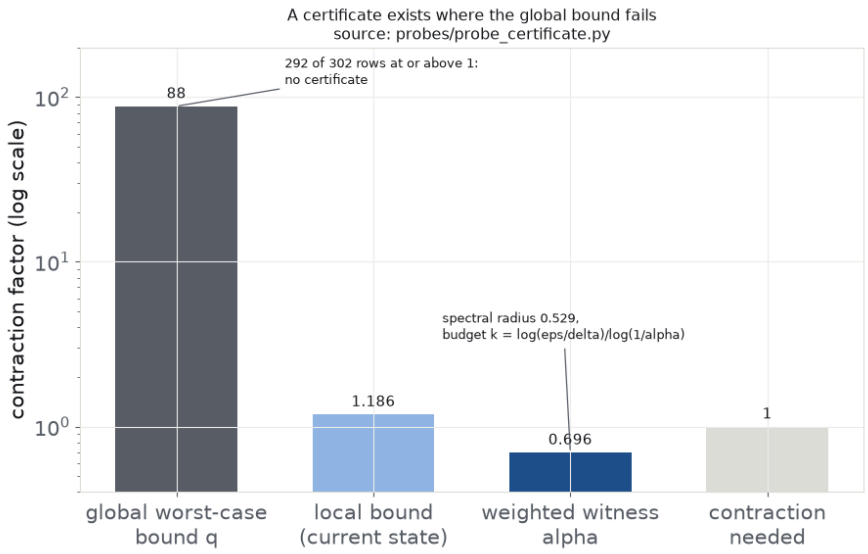


Figure 2: The global worst-case bound, the local bound, the weighted witness, and the unit threshold, on a logarithmic axis. The gap between the first two bars is the price of a state-independent guarantee.

Table 3: Two variants, 1200 ticks each, while the world increases the number of context levels that must be told apart. Neither variant kept a growth. Source: `ep4nn/grow.py`.

variant	kept	rolled back	refused, no response	neurons at the end
without prerequisite gate	0	30	—	302
with prerequisite gate	0	0	1200	302

maximised readout response pinned every answer to the same trajectory, and a rollback that rebuilt the connectome from truncated arrays silently carried the wrong efficacy. The fixes are a response gate that retains a calibration only when it raises the spread, and a snapshot of the connectome together with the efficacy array.

5.4 The prerequisite scan, a negative result

The best configuration moves the readout by 0.0777 across situations while leaving every pair of context answers at a cosine of 1.000: the differences that exist lie along one direction, so the answer has rank one with respect to the situations that must be distinguished. Several configurations collapse to an answer that is identical for all nine situations to three decimals, which indicates a fixed point dominated by the recurrent dynamics rather than by the input. We report this as a negative result about the pair of a measured worm wiring and a rate model at a declared operating point, and not as a claim about connectome substrates in general. The remedy has three candidates in ascending cost: a different declared readout, a dynamics with heterogeneous cells and per-type parameters, or a gain selected by protocol against held-out physiological facts, which is how the library repairs the same failure in the fly antennal lobe.

Table 4: Sixty operating points, ranked by readout spread: five row sums, three class-gain choices, four gain scales, nine situations. Collision is the largest absolute cosine between two situations that share the present input and differ only in context. Source: `probes/probe_readout_response.py`, full data in `docs/assets/prereq_scan.json`.

rank	row sum	class gain	scale	spread	collision
1	0.5	command	10	0.0777	1.0000
2	1.0	command	2	0.0559	0.9999
3	1.0	none	1	0.0328	0.9998
4	0.2	interneuron	10	0.0314	0.9999
5	0.5	command	2	0.0084	0.9999
60	2.0	interneuron	10	0.0000	1.0000

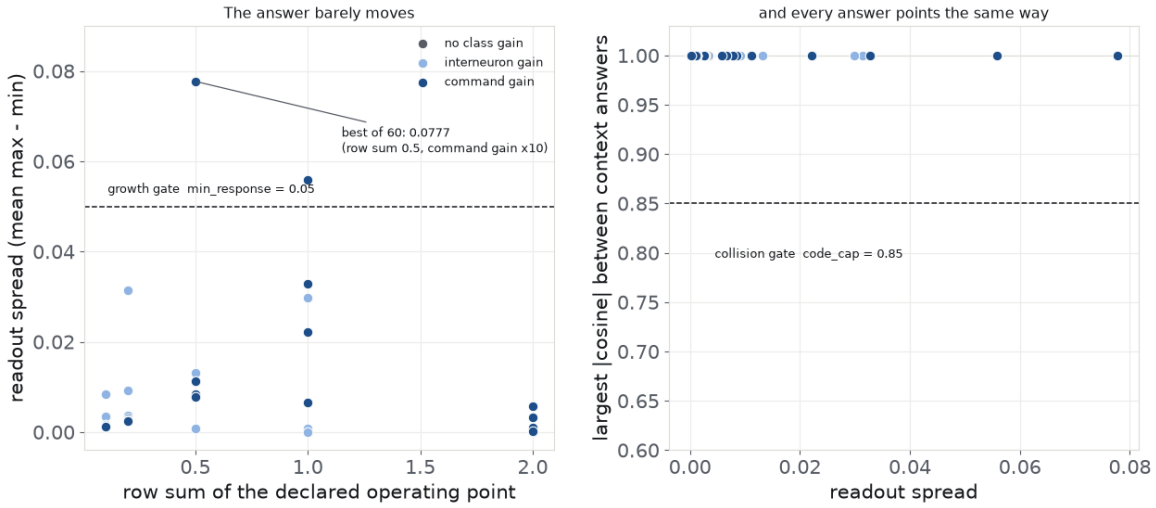


Figure 3: The prerequisite scan. Left: readout spread against the declared row sum for the three class-gain choices, with the growth gate drawn. Right: the largest absolute cosine between context answers against the spread, with the collision gate drawn. Every point sits on the upper shelf, which is why the signal fires on everything and therefore names nothing.

5.5 A negative control result for continuous tracking

The structural reason is the mismatch between the credit window and the plant: the composed brain's eligibility span is 12 steps, while the closed-loop time constant of the plant is 50 to 100 steps. This is consistent with the library's own control-oriented demonstration, which learns a consequence model and leaves the controller supplied. It is also the reason the growth layer is not a remedy for control: a policy that collapses to one action does not present the collisions that growth is keyed to.

6 Discussion

What is shown. The growth contract is implementable on a measured connectome at negligible rebuild cost, its gates fire in the intended order, and the prerequisite measurement is what decides whether structural work is meaningful. Two library-level results are independent of this system:

Table 5: Closed loop on a second-order plant, five force levels, 6000 ticks per arm. The learned arms reduce mean error by about 13 percent, never beat the frozen baseline on the in-band metric, and collapse to a single action level. Source: `probes/probe_control_plant.py`.

arm	mean $\|e\|$	last quarter	in-band, last quarter	dominant action
uniform random	1.020	1.033	0.000	uniform
frozen greedy	1.011	1.067	0.467	1.00
learning, sparse reward	0.880	0.855	0.449	0.824
learning, dense reward, seed 5	0.923	0.919	0.209	0.705
learning, dense reward, seed 11	0.856	0.815	0.459	0.843
learning on every outcome	1.016	1.007	0.000	0.980

a state-dependent certificate exists where the global bound fails, and a sparse graph is settled at about 1.8 ns per synapse, which makes 10,000 neurons a real-time-scale object rather than a research curiosity.

What is not shown. No growth has been kept, and the prerequisite that blocks it is measured rather than assumed. The negative result is a statement about a pair, not about connectomes: it says that the declared readout of this wiring under this model at these operating points does not carry the distinction, which is a stronger and more useful statement than a failed attempt to grow.

Growth is not the same problem as capacity sharing. Specialised small circuits handle single tasks, and the library's own connectome lane reports that a reduced graph matched on population means did not match on member behaviour. Retaining transfer while growing requires shared structure, which is where interference appears. The evidence gate and the outcome gate bound the damage of a wrong growth but do not solve sharing.

Value comes from somewhere. A reward channel needs a scale. A scale supplied by the task caps the system at that task; a scale shaped by selection is endogenous but drifts toward whatever is measured. Both are legitimate designs and both must be declared, because the choice determines which measurements are meaningful. The systems that avoid a human score entirely, by making survival and energy the only currency, are the only ones in this space whose objective is not supplied by a person.

Rule families do not mix. A directed measured connectome does not satisfy the equilibrium contrast gradient contract. Structural growth on directed wiring therefore belongs with a predictive-coding or inference-learning rule, or it must accept that its updates are heuristic. Mixing the two under one set of convergence assumptions is not a defensible default, and the prerequisite gate is what keeps a system honest when the rule family it is using does not apply.

7 Reproducibility

Every table above is produced by one file in `probes/`, and `ep4nn/grow.py` runs its own assertions: nothing shrank, every synapse kept an efficacy across every rebuild, and no kept growth went unmeasured. The environment is a virtual environment with NumPy and SciPy installed from a mirror, and the library is a local checkout with `PYTHONPATH` set. The worm connectome is committed with the repository and its hash recorded. No GPU, no external service and no wallet is involved. Settlement numbers depend on the host, so the receipts record operation counts alongside wall-clock times. The figures of this paper are regenerated from the probe artefacts and the scan dump by one command, and the exact input file of every figure is named in its caption.

8 Conclusion

A measured connectome gives structure cheaply and dynamics not at all. The useful question for a system that grows is not how many neurons it has but whether the answer it produces separates the situations that must be separated. The system presented here encodes that question as a prerequisite, measures it before it grows, and records a refusal to grow as a result in its own right. On the 302-neuron worm graph the prerequisite fails, and reporting that failure is more useful than adding structure to a code that carries nothing. The contract, the receipt format and the state-dependent certificate are the parts we expect to transfer to the next substrate, in particular to connectomes whose motor neurons exist and whose dynamics are heterogeneous.

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